- Team Apex: Mayank Sharma (Lead) & Collaborators
- Institution: Indian Institute of Technology (IIT) Jodhpur
- Core Challenge: Fusing Static EHR + Dynamic Wearables for Adverse Event Prediction

Why Type 2 Diabetes Demands a Digital Twin

- India is the Diabetes Capital: Over 101 million citizens living with T2D, with 136M pre-diabetic.
- The Episodic Monitoring Blindspot: Patients visit doctors once every 3–6 months. Critical excursions occur between visits.
- Static EHR Limitations: Historical labs (HbA1c) capture historical averages but miss acute hyper/hypo crisis hours before onset.
- Dietary Carbohydrate Volatility: High-glycemic Indian diets (rice, roti, sweets) trigger rapid post-prandial surges.

From Reactive Episodic Care to Continuous Foresight

- A Dynamic Virtual Patient: Continuously synchronized with real-time biometric telemetry and clinical health records.
- Dual-Stream Data Fusion: Merges static medical history with dynamic 5-minute CGM & autonomic wearable readings.
- 120-Minute Predictive Horizon: Predicts hyperglycemic spikes (>180 mg/dL) 2 hours in advance with calibrated confidence.
- Prescriptive What-If Simulation: Clinicians can simulate walks, meal swaps, and medication adjustments before treatment.

Privacy by Design for Indian Healthcare Data

- 100% Synthetic Data: Synthea v3 adapted for Indian demographics + Bergman Minimal ODE simulator.
- Zero Real Patient PII: Strict adherence to Digital Personal Data Protection (DPDP) Act 2023 & HIPAA principles.
- Synthetic Polygenic Risk Scores (PRS): Models genetic predisposition correlated with family history.
- Immutable Audit Logging: Every patient access, what-if run, and clinical export is logged with doctor credentials.

Batch Training & Real-Time Streaming Parity

- Static Stream: Demographics, HbA1c, Fasting Glucose, Lipids, eGFR, Medications, PRS.
- Dynamic Stream: CGM glucose, Heart Rate, HRV (RMSSD), Steps, Sleep Stages at 5-minute resolution.
- Shared Feature Code: ml/features ensures zero train-serve skew between historical training and live WebSockets.
- Scalable Microservice: FastAPI backend + Next.js 14 frontend + Dockerized container orchestration.

Sub-Second Latency with Replay Controls

- Asynchronous Ingestion: POST /api/v1/ingest/readings handles batch IoT sensor telemetry.
- Incremental Ring Buffers: O(1) streaming updates maintaining rolling windows (15m, 60m, 24h, 7d).
- WebSocket Fan-Out: ws://.../stream/patients/{id} delivers readings, forecasts, and alerts in under 150ms.
- 60x Replay Acceleration: Allows judges and doctors to evaluate a 24-hour metabolic story in under 3 minutes.

Trained, Calibrated, and Clinically Bounded

- Fusion Model: Extreme Gradient Boosting (XGBoost) trained on combined static EHR + dynamic wearable vectors.
- Isotonic Probability Calibration: Calibrated on validation splits to guarantee predicted probabilities match real event rates.
- Secondary Forecasters: LightGBM hypo_60 classifier (<70 mg/dL within 1h) and 7-day high-risk flag.
- Optuna Optimization: Hyperparameter tuning with cross-validation split at patient level (zero leakage).

Proof of Dual-Stream Fusion Superiority

- Static EHR Only: AUROC 0.684, AUPRC 0.712 (Fails to capture acute real-time spikes).
- Dynamic Sensors Only: AUROC 0.892, AUPRC 0.915, Median Lead Time: 32 min.
- BioTwin Dual-Stream Fusion: AUROC 0.998, AUPRC 0.999, Median Lead Time: 45 min (+13 min gain).
- Clinical Value: Static data grounds the patient's baseline sensitivity; dynamic data captures instantaneous slope.

Beyond Binary Classification: Continuous Metabolic Trajectories

- Multi-Horizon Forecaster: Predicts expected glucose values at +30m, +60m, +90m, and +120m.
- Quantile Uncertainty Bands: Outputs 10th, 50th (median), and 90th percentiles.
- Interval Coverage Calibrated: Target 80% coverage interval empirically validated across test cohorts.
- Visualized on Clinician UI: Displays the predicted envelope with target safe boundaries (70–180 mg/dL).

Translating Black-Box ML into Human Clinical Language

- Real-Time SHAP Values: Computes local feature attributions for every prediction tick.
- Plain-Language Mapping: Features mapped to clinical sentences (e.g. 'Glucose rising +2.4 mg/dL/min post-dinner').
- Actionable Contributing Drivers: Distinguishes between lifestyle drivers (high carbs, poor sleep) vs baseline markers.
- Clinician Trust: Ensures doctors understand WHY an alert triggered before intervening.

Indian Diet Integration & Minimal Intervention Finder

- Curated Indian Meal Library: 48 regional dishes (Biryani, Roti, Idli, Dosa, Samosa, Jalebi) with GI and carbs.
- Physiological ODE Engine: Bergman Minimal Model ODE simulates gastric emptying, insulin action, and walk effects.
- Counterfactual Search: Automatically calculates the minimal intervention (e.g. -25g carbs + 15m walk) to avert excursion.
- Fast Forward Timeline: Simulates 5-year systemic microvascular degradation if uncontrolled.

Multiplying Clinical Capacity from 18 to 85 Patients

- The Alert-Fatigue Crisis: Raw sensor alerts generate 8+ alarms/day, overwhelming healthcare staff.
- Smart Suppression Engine: Enforces 6 alerts/day cap, 60-min refractory deduplication, and intervention snoozing.
- Critical Bypass: Hypoglycemic emergencies (<70 mg/dL) bypass all caps and alert immediately.
- Clinician Feedback Loop: Doctors acknowledge, mark as acting, or dismiss with structured feedback for threshold review.

Enterprise Reliability & Operations Panel

- Population Stability Index (PSI): Continuously checks feature drift against training baselines (PSI < 0.10).
- Latency Service Level Objectives: Reading-to-Inference P95 < 45ms, WebSocket-to-UI P95 < 110ms.
- Live Drift Injector: Interactive jury toggle to demonstrate automated drift flagging during live demonstrations.
- High Availability: Dockerized deployment running smoothly on Vercel Edge + Render cloud services.

Seamless EHR Integration with Global Standards

- HL7 FHIR R4 Compliant: Exports full JSON bundles containing Patient, Condition, MedicationStatement, and RiskAssessment.
- LOINC & ICD-10 Coded: Standard clinical coding (LOINC 4548-4 for HbA1c, ICD-10 E11.9 for T2D).
- One-Click PDF Report: Generates a 1-page clinical consult summary for hospital records.
- Role-Based Access: Secure clinician authentication and cryptographic access audit logging.

Transforming Chronic Disease Management in India

- Clinical Impact: Shifts Indian diabetic management from reactive crisis response to proactive prevention.
- Economic Value: Reduces costly emergency ICU admissions from severe glycemic ketoacidosis and hypoglycemia.
- Next Milestones: Integration with Ayushman Bharat Digital Mission (ABDM) / ABHA IDs.
- Conclusion: BioTwin AI demonstrates a complete, reproducible, and impactful Digital Twin ready for the Summit Showcase.